

```
package com.example.jalsanchaytracker
```

```
import android.os.Bundle
```

```
import androidx.activity.ComponentActivity
```

```
import androidx.activity.compose.setContent
```

```
import androidx.compose.foundation.background
```

```
import androidx.compose.foundation.layout.*
```

```
import androidx.compose.foundation.shape.RoundedCornerShape
```

```
import androidx.compose.material3.*
```

```
import androidx.compose.runtime.*
```

```
import androidx.compose.ui.Alignment
```

```
import androidx.compose.ui.Modifier
```

```
import androidx.compose.ui.graphics.Color
```

```
import androidx.compose.ui.text.font.FontWeight
```

```
import androidx.compose.ui.unit.dp
```

```
import androidx.compose.ui.unit.sp
```

```
class MainActivity : ComponentActivity() {  
    override fun onCreate(savedInstanceState: Bundle?) {  
        super.onCreate(savedInstanceState)  
        setContent {  
            JalSanchayTrackerApp()  
        }  
    }  
}
```

```
@Composable
```

```
fun JalSanchayTrackerApp() {  
    var roofArea by remember { mutableStateOf("") }  
    var rainfall by remember { mutableStateOf("") }  
    var tankCapacity by remember { mutableStateOf("") }
```

```
var harvestedWater by remember { mutableStateOf(0.0) }  
var percentage by remember { mutableStateOf(0f) }  
var message by remember { mutableStateOf("") }
```

```
MaterialTheme {  
    Column(  
        modifier = Modifier  
            .fillMaxSize()  
            .background(Color(0xFFE3F2FD))  
            .padding(20.dp),  
        horizontalAlignment = Alignment.CenterHorizontally  
    ) {
```

```
        Text(  
            text = "Jal-Sanchay Tracker",  
            fontSize = 28.sp,  
            fontWeight = FontWeight.Bold,  
            color = Color(0xFF0D47A1)  
        )
```

```
        Text(  
            text = "Rainwater Harvesting Monitor",  
            fontSize = 16.sp,  
            color = Color.DarkGray  
        )
```

```
        Spacer(modifier = Modifier.height(25.dp))
```

```
        OutlinedTextField(  
            value = roofArea,
```

```
        onChange = { roofArea = it },  
        label = { Text("Roof Area in sq.m") },  
        modifier = Modifier.fillMaxWidth()  
    )
```

```
Spacer(modifier = Modifier.height(12.dp))
```

```
OutlinedTextField(  
    value = rainfall,  
    onChange = { rainfall = it },  
    label = { Text("Rainfall in mm") },  
    modifier = Modifier.fillMaxWidth()  
)
```

```
Spacer(modifier = Modifier.height(12.dp))
```

```
OutlinedTextField(  
    value = tankCapacity,  
    onChange = { tankCapacity = it },  
    label = { Text("Tank Capacity in Litres") },  
    modifier = Modifier.fillMaxWidth()  
)
```

```
Spacer(modifier = Modifier.height(20.dp))
```

```
Button(  
    onClick = {  
        val area = roofArea.toDoubleOrNull()  
        val rain = rainfall.toDoubleOrNull()  
        val tank = tankCapacity.toDoubleOrNull()
```

```

        if (area == null || rain == null || tank == null || tank <= 0) {
            message = "Please enter valid numbers."
            harvestedWater = 0.0
            percentage = 0f
        } else {
            val runoffCoefficient = 0.8
            harvestedWater = area * rain * 0.929 * runoffCoefficient
            percentage = (harvestedWater / tank).toFloat().coerceIn(0f, 1f)

            message = "You can harvest %.2f litres of water.".format(harvestedWater)
        }
    },
    modifier = Modifier.fillMaxWidth(),
    colors = ButtonDefaults.buttonColors(Color(0xFF1976D2))
) {
    Text("Calculate Water Saved")
}

Spacer(modifier = Modifier.height(25.dp))

WaterTankView(percentage)

Spacer(modifier = Modifier.height(20.dp))

ResultCard(
    title = "Water Saved Report",
    content = if (message.isEmpty()) {
        "Enter roof area, rainfall and tank capacity to calculate water saving."
    } else {
        message + "\nTank Filled: ${((percentage * 100).toInt())}%"
    }
}

```

```
)
```

```
Spacer(modifier = Modifier.height(16.dp))
```

```
TipsCard()
```

```
}
```

```
}
```

```
}
```

```
@Composable
```

```
fun WaterTankView(percentage: Float) {
```

```
Card(
```

```
    modifier = Modifier
```

```
        .width(180.dp)
```

```
        .height(260.dp),
```

```
    shape = RoundedCornerShape(24.dp),
```

```
    colors = CardDefaults.cardColors(Color.White),
```

```
    elevation = CardDefaults.cardElevation(8.dp)
```

```
) {
```

```
Box(
```

```
    modifier = Modifier
```

```
        .fillMaxSize()
```

```
        .padding(15.dp),
```

```
    contentAlignment = Alignment.BottomCenter
```

```
) {
```

```
Box(
```

```
    modifier = Modifier
```

```
        .fillMaxWidth()
```

```
        .fillMaxHeight(percentage)
```

```
        .background(
```

```
            Color(0xFF42A5F5),
```

```

        RoundedCornerShape(18.dp)
    )
)

Text(
    text = "${(percentage * 100).toInt()}%",
    fontSize = 26.sp,
    fontWeight = FontWeight.Bold,
    color = Color(0xFF0D47A1),
    modifier = Modifier.align(Alignment.Center)
)
}
}
}

```

@Composable

```

fun ResultCard(title: String, content: String) {
    Card(
        modifier = Modifier.fillMaxWidth(),
        shape = RoundedCornerShape(18.dp),
        colors = CardDefaults.cardColors(Color.White),
        elevation = CardDefaults.cardElevation(6.dp)
    ) {
        Column(modifier = Modifier.padding(16.dp)) {
            Text(title, fontSize = 20.sp, fontWeight = FontWeight.Bold)
            Spacer(modifier = Modifier.height(8.dp))
            Text(content, fontSize = 15.sp)
        }
    }
}
}

```

```

@Composable
fun TipsCard() {
    Card(
        modifier = Modifier.fillMaxWidth(),
        shape = RoundedCornerShape(18.dp),
        colors = CardDefaults.cardColors(Color(0xFFE8F5E9)),
        elevation = CardDefaults.cardElevation(4.dp)
    ) {
        Column(modifier = Modifier.padding(16.dp)) {
            Text(
                text = "Water Saving Tips",
                fontSize = 20.sp,
                fontWeight = FontWeight.Bold,
                color = Color(0xFF1B5E20)
            )

            Spacer(modifier = Modifier.height(8.dp))

            Text("• Clean roof and pipes before rainy season.")
            Text("• Use filters before storing rainwater.")
            Text("• Reuse saved water for gardening.")
            Text("• Check tank overflow regularly.")
            Text("• Save every drop for future use.")
        }
    }
}

```

Formula used

Harvested Water = Roof Area × Rainfall × 0.929 × Runoff Coefficient

This code includes:

Rainfall input

Roof area input

Tank capacity input

Water saved calculation

Digital tank filling animation

Monthly report style result

Tips section

Error handling for wrong input